

SHIVAM YADAV

FULL STACK DEVELOPER

Agra, Uttar Pradesh 282002, India | shivam_y@bt.iitr.ac.in | +91 6397556533 | [LinkedIn](#)

PROFESSIONAL EXPERIENCE

Full Stack Development Intern, GPT Innovations

May 2025 to Aug 2025

- Built responsive Next.js React components with optimized server-side rendering, significantly improving UI performance and user experience for healthcare providers managing high-volume billing and claims operations across the platform.
- Integrated backend APIs seamlessly with AI modules including LLMs, LangChain, OCR, and NLP pipelines to automate document processing, clinical documentation, prior authorization, and payer workflows across the full healthcare revenue cycle platform.
- Built modular MCP servers on n8n.io by converting automation workflows into reusable control-plane components for AI.
- Designed and deployed HIPAA-compliant LLM-powered voice agents and speech-processing pipelines using LangChain to fully automate 24/7 multilingual patient scheduling, payer calls, and prior authorization workflows across all healthcare operations.
- Directed end-to-end integration across frontend, backend, and AI services to improve reliability and reduce manual handoffs.

EDUCATION

Indian Institute of Technology Roorkee

Oct 2022 to May 2026

B.Tech. in Biosciences and Bioengineering, Roorkee, Uttarakhand

GPA: 7.303/10

Milton Public School

Apr 2019 to Apr 2020

Intermediate (Class XII), Agra, Uttar Pradesh

Percentage : 84.60

Gayatri Public School

Apr 2017 to Apr 2018

Matriculate (Class X), Agra, Uttar Pradesh

Percentage : 86.66

PROJECTS

CareLink RBAC Healthcare Platform , Department of Biosciences and Bioengineering

Oct 2024 to Sept 2025

- Engineered an RBAC healthcare platform for IIT Roorkee's Biosciences Department, streamlining multi-hospital workflows for Admins, Doctors, and Hospital Admins to ensure secure, role-based, end-to-end patient data management across departments.
- Designed modules for hospital admins to manage institutional profiles and for doctors to log treatment updates, observations, and scheduled entries — enabling long-term longitudinal tracking of detailed patient health records across hospitals.
- Built an interactive React dashboard with PostgreSQL APIs for real-time monitoring of patient recovery and treatment progress.
- Architected the data layer to support upcoming ML models for accurate burn classification, laying the foundation for AI-assisted diagnostics, clinical decision-making, advanced treatment planning, and longitudinal recovery tracking across hospital settings.

SMThub Protein-DNA Binding Visualization Web Server, Department of Biosciences and Bioengineering

Aug 2025 to Dec 2025

- Developed a bioinformatics web server for visualizing protein-DNA binding footprints from bisulfite sequencing data.
- Designed algorithms mapping bisulfite sequencing data into three chromatin states for transcription factor detection.
- Integrated R (Bioconductor) with a Node.js backend to perform reproducible statistical computation on raw genomic data, bridging biological analysis tools with modern web infrastructure to support scalable cross-organism genomic research.
- Built a React frontend with PostgreSQL APIs for dynamic queries, interactive heatmap visualizations, and exportable outputs.

RESEARCH PUBLICATIONS

- Aashna Bansal, Himani Barmola, **Shivam Yadav**, Satyanarayan Rao, "SMTrackR: a R/Bioconductor package for mapping protein binding at individual DNA molecules", in Software Package, bioconductor.org, 2026
- K. C. Ananth Kumar, Adwaita Nair, Shreya Sharma, Deepika Singh, **Shivam Yadav**, Devesh Bhimsaria, Sebanti Gupta & Saugata Hazra, "Strategic Design of a Multi-Tier Database for Class A -Lactamase BlaC Variants of M. tuberculosis: Advancing the Fight Against Antibacterial Resistance", in The Journal of Antibiotics, Springer Nature, 2025

TECHNICAL PROFICIENCIES

Skills : C++, JavaScript, TypeScript, Python, SQL
Frameworks: Next.js, Express.js, React.js, Prisma, Tailwind CSS
Databases: PostgreSQL, MongoDB, Redis
Tools & Cloud: Git, Github Actions, Docker, Postman, AWS, n8n